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Lithium Potential in Saudi Arabia – An Initial Assessment

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Abstract

Lithium (Li) is a key element in the global energy transition, playing a vital role in low-carbon energy storage systems, particularly in batteries for electric vehicles (EVs). This study provides the first comprehensive geological assessment of lithium potential in Saudi Arabia by examining the natural occurrence of lithium in igneous and sedimentary rocks and the processes that lead to its enrichment in subsurface brines. The results suggest that the west coast rift basins and the Arabian Shield hold significant potential, while the eastern sedimentary basin appears to be less favorable.

Lithium naturally occurs in felsic igneous rocks, especially in their glass and micaceous components, and in sedimentary rocks derived from them, such as micaceous sandstones. Enrichment of lithium in pore waters may result from two primary mechanisms: (1) Water–rock interaction at elevated temperatures (>120°C), which leaches lithium from host minerals into formation fluids; and (2) Evaporative concentration of seawater, where lithium remains in the residual brine during progressive evaporite precipitation.

The felsic igneous rocks of the Arabian Shield and syn-rift sedimentary formations along the Saudi west coast show strong potential as lithium sources. In addition, evaporative processes in the region that lead to the precipitation of thick salt deposits further contribute to lithium enrichment in subsurface brines. Although current data is limited, the study identifies the west coast rift basins as the most promising areas for lithium-rich deep brines suitable for Direct Lithium Extraction (DLE). Lithium originates from the leaching of shield sediments in the deep rift basins and from leftover brines associated with the km-thick Miocene evaporates deposited in the Red Sea rift. In contrast, the Eastern Arabian Basin shows limited potential for lithium-rich brines. Evaporites are limited to a thin gypsum layer in the Jurassic Hith and older (Precambrian) deeply buried Hormuz salts, while clastic material eroded from the shield is shallow and too proximal to the outcrops for significant contributions to basinal brines.

This study provides the first comprehensive assessment of lithium potential in Saudi Arabia, offering valuable insights into sustainable resource development and future exploration opportunities in the Kingdom.

Introduction

Lithium is a light metal that plays an important role in the energy transition. It has an atomic number of three (3) and is highly reactive. It can be found in different chemical compounds such as Lithium Chloride (LiCl), Lithium carbonate (Li_2CO_3), Lithium Hydroxide (LiOH), and others. It was previously used mainly in the ceramic and glass industries to give high mechanical strength and thermal shock resistance; however, due to its low electrical resistivity and light weight, it is ideal for use in batteries (British Geological Survey, 2016). Lithium-ion batteries have high energy density, meaning that they last longer and are lighter than other batteries. These types of batteries are used in different devices, such as cellphones, laptops, and electric vehicles (EVs), the latter of which is essential for the goal of a low-carbon transportation and energy system.

In order to reach a low-carbon energy system, the electrification of transport is needed; because of this, the demand for Lithium is expected to increase (BP p.l.p, 2024). However, the traditional commercially exploitable reserves are limited, with their extraction being highly damaging to the environment. Additionally, many are geographically restricted; for example, the evaporation ponds or salars, such as the salars in Argentina, Chile, Bolivia, and China that occur in the arid latitudinal belts between 19° and 37° north or south (Fig. 1, (Bradley et al., 2013)). Due to the increase in the lithium demand, there is a need for other countries to assess their own lithium resources to ensure their supplies for the transition (Gourcerol et al., 2019). One option that has started to attract attention is the extraction of lithium from deep geothermal brines (e.g., Salton Sea, USA (Mckibben et al., 2021)). Extracting lithium from geothermal brines has a smaller local environmental impact than traditional production types. For instance, mining requires blasting and sulfuric acid digestion, and evaporation ponds require large areas of land and significant quantities of fresh water, in addition to the carbon footprint of transporting these lithium products from overseas (Mckibben et al., 2021).

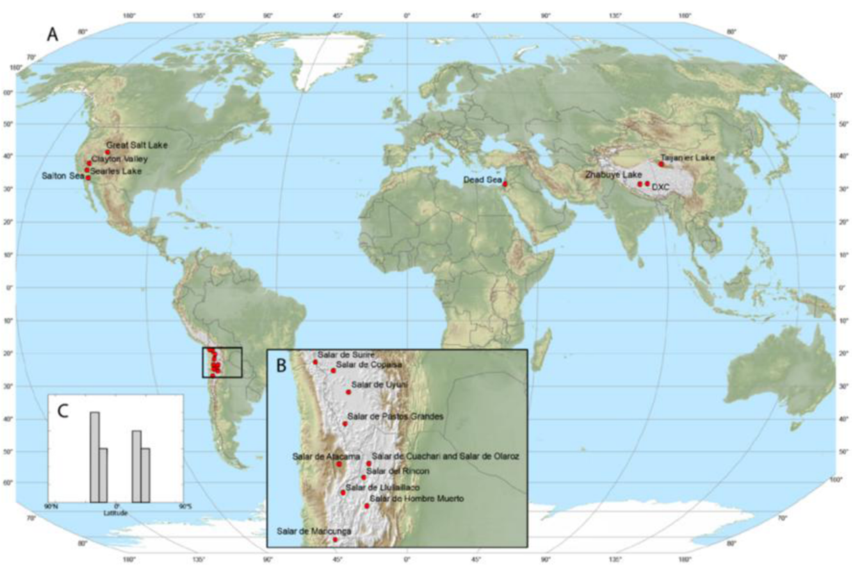

Figure 1—Evaporation ponds or salars are geographically restricted; they occur in the arid latitudinal belts between 19° and 37° north or south. (a) World map of evaporation ponds or salars (red dots). (b) Zoom in on South America. (c) Histogram showing the bimodal latitudinal distribution of evaporation ponds or salars in northern and southern arid belts. From (Bradley et al., 2013).

Some geothermal brines around the world are strongly enriched in lithium, containing up to 500 ppm of this metal (Moldovanyi & Walter, 1992; Sanjuan et al., 2020, 2022). To locate subsurface brines with similar composition in the Kingdom of Saudi Arabia, it is necessary to understand the possible provenance of lithium and the processes behind its enrichment in brines.

Lithium in minerals and brines

Lithium production types

Three different lithium production types exist (Sanjuan et al., 2022): (1) Surface or near-surface brines, also called evaporation ponds or salars (Fig. 2a). This production type occurs mainly in Chile, Argentina, and Bolivia, known as the Lithium Triangle, and in China. These salars produce lithium carbonate (Li_2CO_3) as their final product through conventional solar evaporation; however, this evaporation process requires large extensions of land, and it is time-consuming (Boroumand & Razmjou, 2024). (2) Hard rock deposits, where lithium is extracted from lithium-rich minerals, such as spodumene and lepidolite (Fig. 2b), which pass through mechanical and chemical treatments, including acid leaching, to produce lithium carbonate and lithium Hydroxide (LiOH) as the final product. Australia, Canada, China, and the United States of America are the main countries with this type of ore. Finally, (3) the emerging alternative resources, where lithium is extracted from seawater or geothermal brines using different technologies for Direct Lithium Extraction (DLE). These alternative resources have a significantly lower environmental impact than the conventional producer types. A current example of lithium extraction from geothermal brines is the Salton Sea project in the United States (Fig. 2c), where different technologies for Direct Lithium Extraction (DLE) are implemented after energy from heat has been extracted.

Figure 2—Lithium producer types. (a) Surface or near-surface brines. Photograph of the Salar de Atacama, Chile mineral (British Geological Survey, 2016). (b) Hard rock deposits. Spodumene from the Walnut Hill pegmatite, Huntington, Hampshire County, Massachusetts, USA (British Geological Survey, 2016). (c) Emerging alternative resources. Geothermal lithium production at the Salton Sea, USA (Cariaga, 2021)

Comparing the two resources included in the emerging alternative production type, the lithium concentration in seawater is, on average, 0.2 ppm, while in geothermal brines, it varies from close to 0 ppm to up to 500 ppm (Table 1). One aspect that could impact the economic feasibility of lithium recovery from fluids is the magnesium-lithium concentration ratio (Mg^{2+}/Li^{+}). These two ions have similar ionic radii, 0.72 Å for Mg^{2+} and 0.69 Å for Li^{+} (Marcus, 1991). This similarity could limit their separation, for example, some membranes are not able to sufficiently separate these two ions, as both will pass through the small membrane sieve.

Table 1—Lithium-rich brines around the world.

Location	*Li content (ppm)*	*Temperature (°C)*	*TDS (g/L)*	*Depth (m)*
*Italy**	*280 - 480*	*>300*	*230 - 516*	*1000 - 3000*
*Germany-France / Upper Rhine Graben**	*160 - 210*	*120 - 250*	*90 - 120*	*2500 - 5000*
*Germany / Molasse Basin**	*140 - 160*	*70 - 210*	*55 -62*	*1950*
*North Germany**	*180 - 237*	*150 -220*	*210 - 260*	*3500 - 4300*
*United Kingdom**	*125*	*52 – 70*	*19*	*690*
*United States / Salton Sea***	*250*	*330*	*260*	*2500 – 3200*
*United States / Smackover****	*10 - 423*	*51 - 104*	*150 - 364*	*1325 - 3250*
*China *****	*20 - 239*	*70*	*1 - 5*	*Hot springs*
*West-Central Alberta, Canada******	*12 - 112*	*-*	*90 - 234*	*2500–3900*

(Sanjuan et al., 2020, 2022)* , (Mckibben et al., 2021)**, (Moldovanyi & Walter, 1992)***, (Wang et al., 2021)****, (Eccles & Berhane, 2011)*****

The Mg^{2+}/Li^{+} ratio in seawater is high, around 7000, while in geothermal brines, it varies from less than 0.001 up to 130. The optimal Mg^{2+}/Li^{+} ratio for economic lithium extraction from fluids is below 10 (Dini et al., 2022) making subsurface brines significantly more favorable for a membrane-based extraction. However, this limiting aspect will also depend on the advancements of different extraction technologies. Currently, adsorption and membrane methods are the two prevailing technologies used for lithium extraction from aqueous solutions.

Lithium occurrence

The average lithium concentration in the crust is around 17 to 20 ppm. However, in some igneous and sedimentary rocks, this concentration can be much higher (more than 4000 ppm) (British Geological Survey, 2016).

Igneous rocks. The lithium content in felsic to intermediate rocks is higher than in mafic and ultramafic rocks (Dini et al., 2022; Sanjuan et al., 2022). The more differentiated the rock, the higher its lithium content. For example, highly fractionated granites (e.g., leucogranites) have lithium concentrations up to 2000 ppm (Wang et al., 2021). This is due to the incompatibility of lithium in the magma system, leading to an accumulation in the residual melt during the magma differentiation process. These felsic rocks include plutonic rocks such as granite, volcanic rocks such as rhyolite, and pyroclastic rocks such as volcanic tuff and ignimbrites with felsic composition.

Lithium in felsic volcanic and pyroclastic rocks

Lithium is not incorporated in the structure of most magmatic minerals. Because of this incompatibility, there is a relative enrichment of lithium in the glass component of the felsic volcanic and pyroclastic rocks (Dini et al., 2022). The glass component is easily leachable, releasing lithium into subsurface fluids by fluid-rock interaction.

Lithium in plutonic rocks

Alternatively, when the igneous rock does not have glass, such as in plutonic rocks, lithium will preferentially be incorporated in biotite and white micas (Sanjuan et al., 2022). These micas are phyllosilicates that have octahedral sites where Mg is usually located. Because of the similarity in radius between Mg and Li, Li may substitute for Mg and form a Li-rich mica. Through water-rock interactions at temperatures higher than 120°C, these micas can be dissolved, and lithium can be released into the fluid (Dini et al., 2022; Sanjuan et al., 2022).

Lithium in hard rock deposits

Among the minerals that contain lithium, only a few are economically extractable. The most abundant lithium-bearing mineral is spodumene ($LiAlSi_2O_6$). This mineral occurs in granites and pegmatites. Other common lithium-bearing minerals are lepidolite ($K_2(Li,Al)_{5\text{-}6}\{Si_{6\text{-}7}Al_{2\text{-}1}O_{20}\}(OH,F)_4$) and petalite (LiAlSi4O10), which occur in pegmatites (British Geological Survey, 2016).

Sedimentary rocks.

Lithium in clays

Some octahedral clay minerals, for example, smectite, illite, and chlorite, are alteration products caused by water-rock interaction in geothermal systems. For example, when the glass component in volcanic rocks is altered, it forms illite-smectite clays (Sanjuan et al., 2022). These octahedral clays can incorporate Li in their structure; therefore, the alteration of Li-rich volcanic glass can produce a Li-rich illite-smectite clay called hectorite. Another way of forming this hectorite is when there is a pre-existing smectite clay deposit, and there is circulation of a fluid containing lithium. The pre-existing smectite clay can adsorb lithium from the fluid, forming a hectorite deposit (British Geological Survey, 2016; Dini et al., 2022).

Lithium in sandstones

Volcano-sedimentary rocks formed by felsic rocks as detrital material, such as micaceous sandstones, might be enriched in Li. Though water-rock interactions at temperatures higher than 120°C, the micas can be dissolved, and lithium can be released into the fluid (Dini et al., 2022; Sanjuan et al., 2022). As an example, in the Li-rich brines located in the Upper Rhine Graben on the border between France and Germany, the primary source of lithium is probably a thick micaceous continental sandstone with a minor contribution of the granitic basement (Sanjuan et al., 2022).

Lithium sources in subsurface brines.

Lithium from fluid-rock interaction processes

The most commonly accepted possible source of lithium in subsurface brines is due to fluid-rock interaction processes with Li-bearing minerals in rocks (Fig. 3). These rocks could be felsic igneous rocks such as granite, rhyolite, and pyroclastic rocks (volcanic tuff and ignimbrites with similar composition), and volcano-sedimentary rocks formed by detrital material derived from felsic rocks. Lithium can be found in the glass component of volcanic rocks or the micas of plutonic rocks. In either case, at high temperatures (>120°C), lithium is leached from the rock and added to the brine.

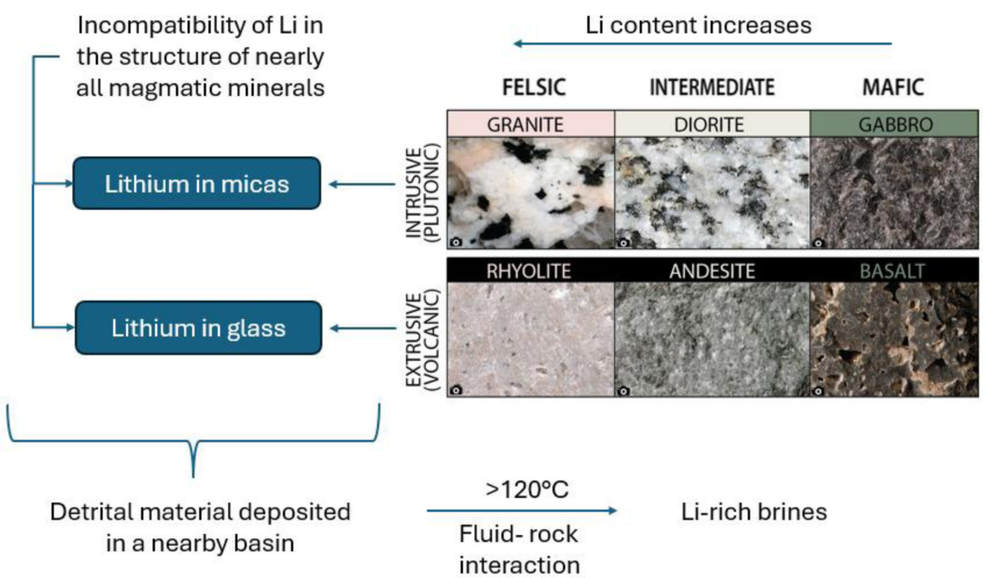

Figure 3—Lithium source through fluid-rock interaction. Image modified from (Panchuk, 2019). The more differentiated the igneous rock, the higher the lithium content. Lithium is not incorporated in most magmatic minerals; therefore, there is a relative enrichment of Li in the glass component for volcanic rocks. In plutonic rocks, Li is hosted in micas (biotites and white micas). Additionally, volcano-sedimentary rocks derived from felsic rocks (e.g., micaceous sandstones) are also sources of Li. Through fluid-rock interaction and temperatures higher than 120°C, glass and micas are leached, and the liberated Li is incorporated into the fluid.

Lithium from seawater evaporation

Another possible source of lithium in subsurface brines is through seawater evaporation, where Li from seawater is concentrated in the remaining fluid while evaporites precipitate (Fig. 4). As seawater evaporates in a closed basin, the concentration of ions in the water increases, and depending on the degree of evaporation, different minerals precipitate.

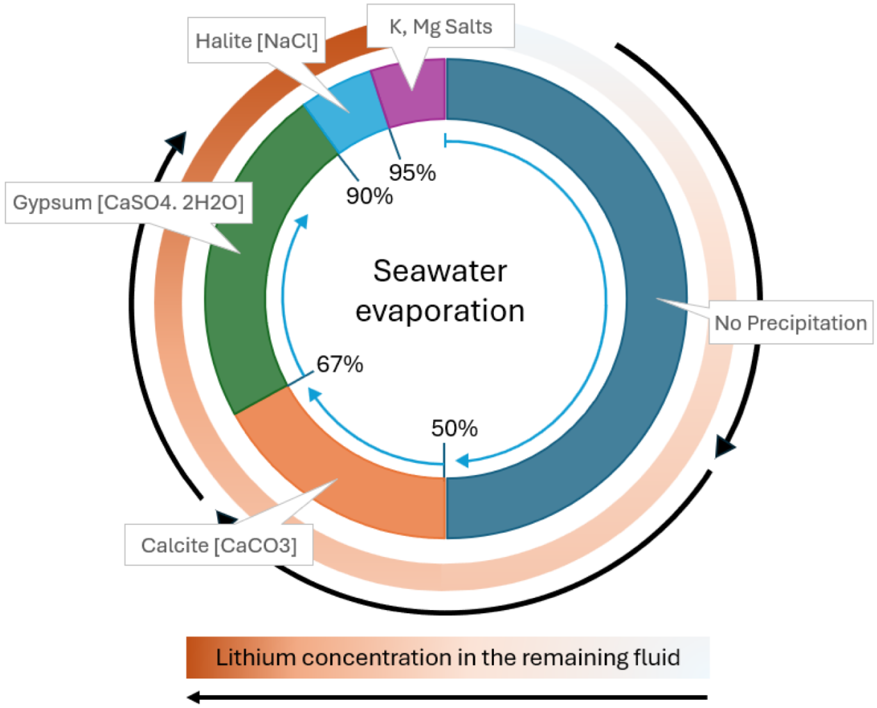

Figure 4—Evaporative brine concentration. Concentration of seawater by evaporation showing minerals that precipitate when seawater is gradually evaporated (after James & Jones, 2016). As seawater evaporates, lithium concentration increases in the fluid. Only K and Mg salts (potash) have a significant amount of Li when they precipitate.

Lithium is highly incompatible with most minerals, including all evaporitic minerals such as calcite, gypsum/anhydrite, halite, and potassium-magnesium salts. As a result, lithium is not directly incorporated into their crystalline structures. For example, Li has a low (<<1) distribution coefficient in halite (Godfrey et al., 2013). For this reason, as seawater evaporates and different minerals precipitate, lithium remains in the concentrated remaining brine. The more evaporation there is, the higher the lithium concentration in the fluid.

Although lithium is not incorporated directly into the structure of evaporites, some Li can be trapped in these minerals as they precipitate; this mechanism is called occlusion. The concentration of a trace element occluded is a function of the precipitation rate and the concentration of the element in the solution (Dean, 1987). For instance, trace elements will increase in the last stages of halite precipitation rather than at the beginning because the concentration of the trace elements is higher in the fluid in the last part of this evaporation-concentration process. Nevertheless, lithium concentration in evaporites is generally low, except for Potash (K and Mg salts). These K and Mg salts can have a Li concentration of up to 20 – 40 ppm, as reported for the main Ethiopian Rift Lake (Bekele & Schmerold, 2020).

Bromine (Br) is another element that behaves similarly to lithium (Li). Only a little Br is incorporated in gypsum or halite. (Moldovanyi & Walter, 1992). For this reason, the evolution of the concentration of major elements during evaporation is often expressed relative to bromide. This fact can help delimit the amount of Li that may come from seawater evaporation. For example, in the brines extracted from a well in the Smackover formation of the United States, which consists of limestones and dolostones, the concentrations of Br and Li are high, up to 4276 ppm and 423 ppm, respectively Table 2. In seawater, the concentrations of Br and Li are around 65 ppm and 0.2 ppm, respectively. Therefore, assuming that the Smackover brines are seawater concentrates and that Br/Li ratios remained unchanged, the lithium content should only be 13.2 ppm. Hence, as suggested by Moldovanyi & Walter, (1992) the observed Li enrichment in Smackover brines most likely results from additional water-rock interactions with older siliciclastic units and subsequent brine mixing with those deeper brines.

Table 2—Concentration of elements in brines

Field	Salton Sea*	Upper Rhine Graben**	Upper Rhine Graben**	Upper Rhine Graben**	Smackover formation***	Smackover formation***
		Sedimentary-Triassic Permian	Granite-Carboniferous	Sedimentary-Oligocene	Jurassic	Jurassic
Temperature	330 °C	120 °C	160 °C	-	84 °C	94 °C
Depth (m)	2500-3220	2542	3044	561	2425	2665
Concentration in ppm						
Na	54800	35127	28200	31100	73330	72500
Ca	28500	7363	7700	233	38674	29300
K	17700	3113	4000	209	2672	7100
Fe	1710	42.38	21.7	4.36	-	
Mn	1500	25.17	26.1	0.614	15	56
SiO2	588	77.4	159	15.8	24	36
Zn	507	15.223	6.3	-	-	
Sr	421	391	430	214	2560	2258
B	271	39.4	39	23.6	187	358
Ba	210	9.287	11.5	70	26	49
Li	209	159	179	17.6	223	423
Mg	49	301	76	1450	3348	2243
NH4	330	47.7	37.2	59.6		
Cl	157500	73566	64200	55100	197405	176679
Br	111	203	219	356	5426	4276
CO2	1580	15	15	39.2		
HCO3	NA	<2	0.7	6		
H2S	10				189	364
SO4	53	267	126	178	166	107
TDS	265 g/L	121 g/L	106 g/L	91 g/L	325 g/L	297 g/L
Mg/Li	0.234	1.893	0.425	82.386	15.013	5.303

(Mckibben et al., 2021)*, (Sanjuan et al., 2022)**,(Moldovanyi & Walter, 1992)***

Additionally, these two possible sources, fluid-rock interaction and seawater evaporation, complement each other in the way that higher salinity, due to greater degrees of seawater evaporation, promotes greater diagenetic reactions (Hanor, 2001), leading to stronger water-rock interactions. Therefore, Li concentration in brines depends on temperature, fluid salinity, and the type of reservoir rock and its mineralogical constituents (Sanjuan et al., 2022).

Lithium in Saudi Arabia

Lithium sources

Lithium from Igneous Sources in the Arabian Shield. The Arabian shield contains several intermediate and felsic plutonic rocks ranging from dioritic to granitic compositions (Drysdall et al., 1986). Some of these rocks contain an associated mineralization and have been geochemically analyzed. For example, in the Jabal Ghadara area in the central Arabian shield, there are felsic to intermediate rocks with associated gold mineralization (location shown in Fig. 5). The chemical analysis in this area shows a Li content of up to 32 ppm for a diorite, which is an intermediate rock, and a Li content of up to 81 ppm for a monzogranite, a felsic rock (Harbi et al., 2016). Another example of felsic rocks in the Arabian Shield are the white and black volcanoes of the Harrat Khaybar, north of Madinah (location in Fig. 5, image in Fig. 6). The white volcanoes are rhyolitic (felsic composition), while the black ones are basaltic (mafic composition) (Kaminski et al., 2014).

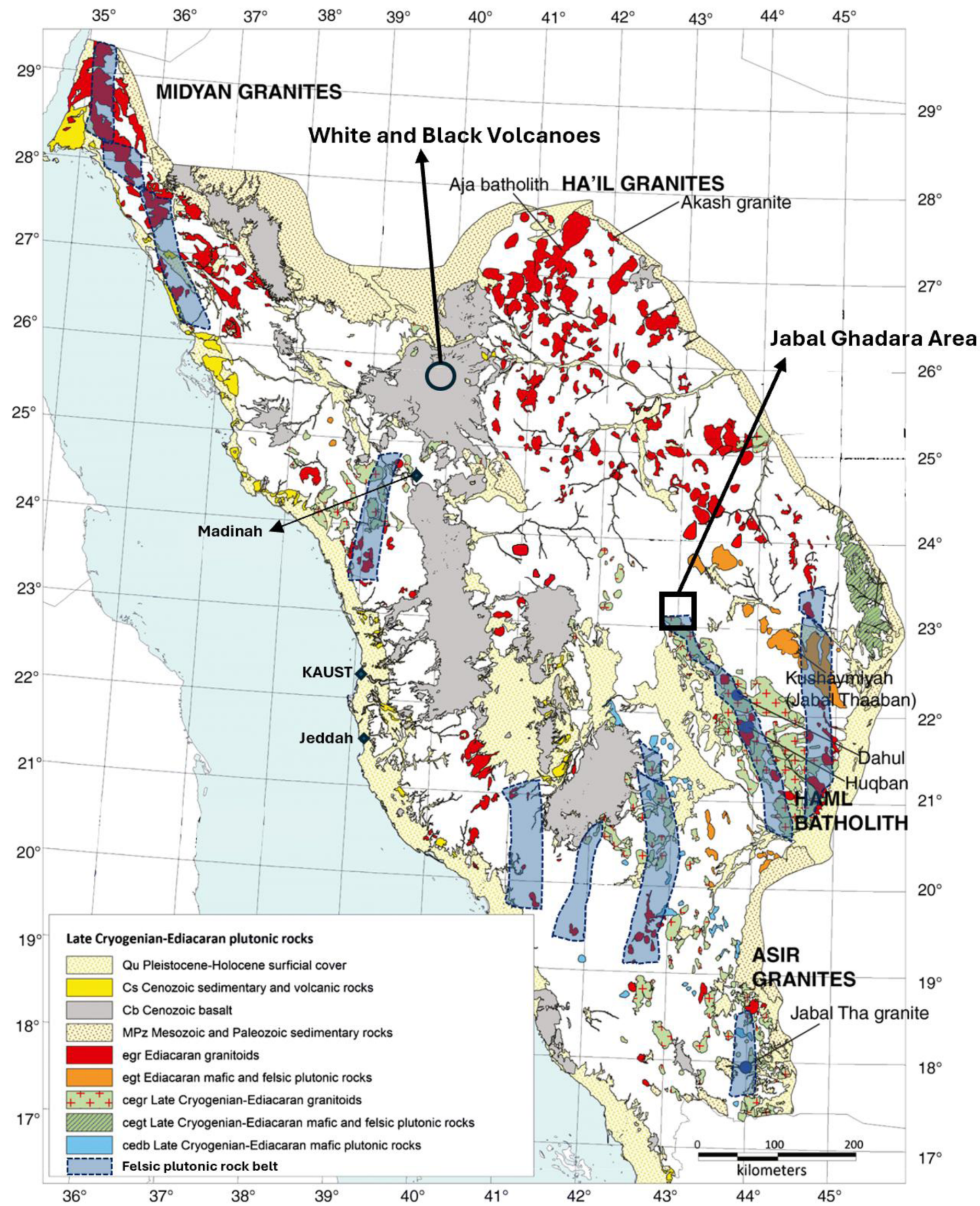

Figure 5—Map showing the distribution of granites in the Arabian Shield (modified from Al-Saleh & Al-Omari, 2021), along with more differentiated felsic plutonic rock belts. (Modified from Ramsay, 1986). The Location of the gold mineralization in Jabal Ghadara is highlighted in the black box (from Harbi et al., 2016), and the location of the White and Black volcanoes is in the black circle.

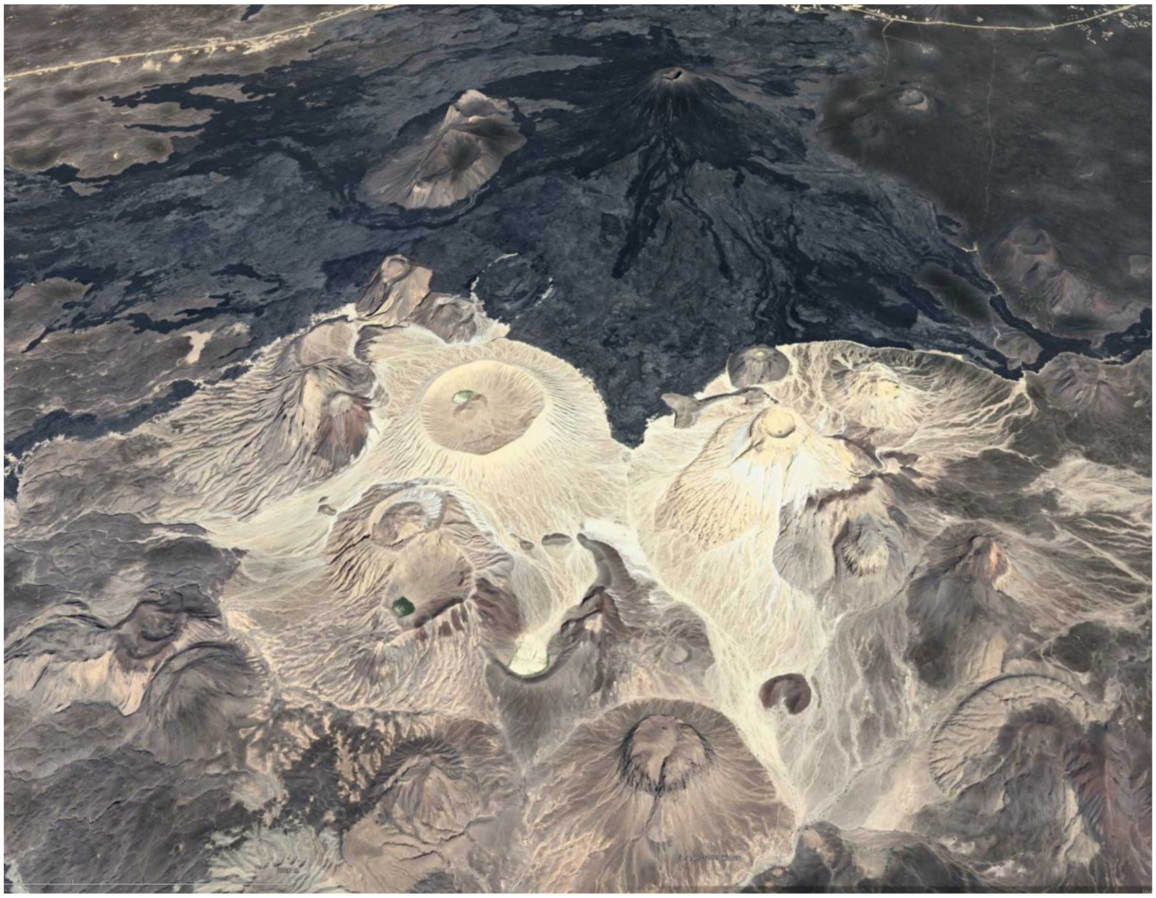

Figure 6—White and black volcanoes of Harrat Khaybar, north of Madinah. Scale Bar 3km. Image Google Earth (15 Feb 2025).

Furthermore, the Arabian shield contains several clusters and belts of felsic plutonic rocks that are reported to contain even higher lithium concentrations of up to 121 ppm; these rocks have formed as an end product of magmatic differentiation, explaining the high concentrations of incompatible trace elements (Ramsay, 1986). The distribution of the granites in the Arabian Shield, along with the more differentiated felsic plutonic rock belts, are shown in Fig. 5.

While there are felsic belts with relatively high lithium concentrations in the shield, reports of Li-bearing mineral deposits suitable for hard-rock mining have yet to emerge, offering exciting potential for future exploration.

Lithium in sedimentary rocks. Sedimentary rocks that contain sediments derived from the erosion of felsic rocks from the shield have the potential to release lithium into the fluid through water-rock interactions. An example of these sedimentary rocks are micaceous sandstones. To enrich the fluid in lithium, the water-rock interactions require elevated temperatures (greater than 120°C) and sufficient time. Considering a geothermal gradient of 30 °C per kilometer and an annual average surface temperature of 30 °C, a burial depth of about 3 km is required to lead to mineral leaching and brine enrichment in lithium.

Large quantities of sediments eroded from the shield were carried westward and deposited in rift basins along the west coast of Saudi Arabia. An example is the Al Wajh formation, which represents the oldest synrift unit in the rift basins (Fig. 7; Hughes & Johnson, 2005; Tubbs et al., 2014). This formation is primarily composed of red-gray mudstones, arkosic sandstones, and conglomerates (Hughes & Johnson, 2005). The detrital composition of the Al Wajh formation is dominated by quartz, with additional contributions from feldspars, micas, and lithic fragments. These lithic fragments include both sedimentary and igneous rocks, with the igneous fragments primarily consisting of plutonic felsic rocks (Bello et al., 2023).

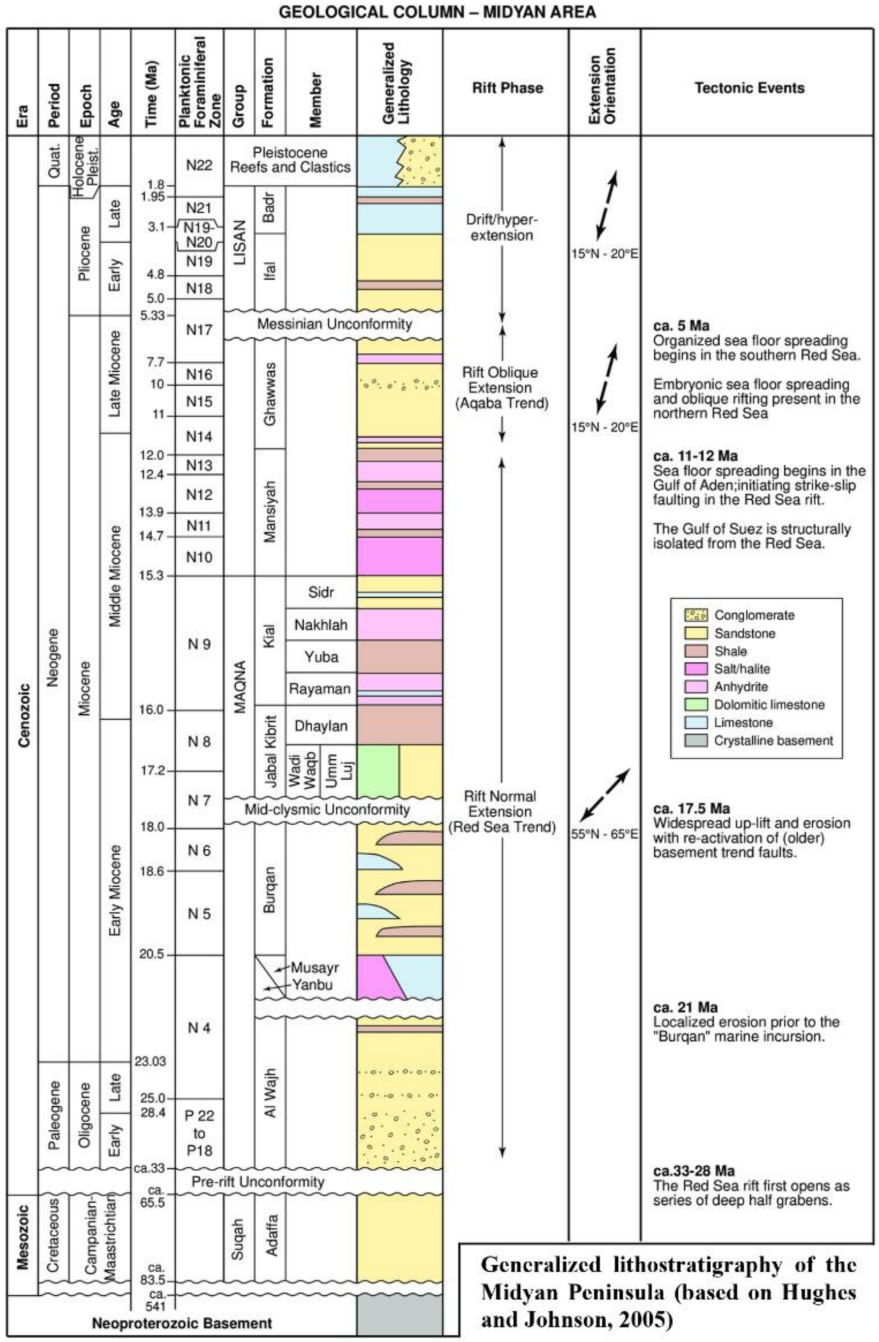

Figure 7—Generalized lithostratigraphy of the Midyan Peninsula (based on Hughes and Johnson, 2005; from Tubbs et al., 2014)

Several cross-sections of the west coast rift basins (Fig. 8, Baby et al., 2024), show areas at the center of the half-grabens where the basins reach significant depths exceeding 5 km. The Al Wajh formation, the oldest synrift unit, extends to these depths, providing the conditions necessary to reach temperatures of over 120°C.

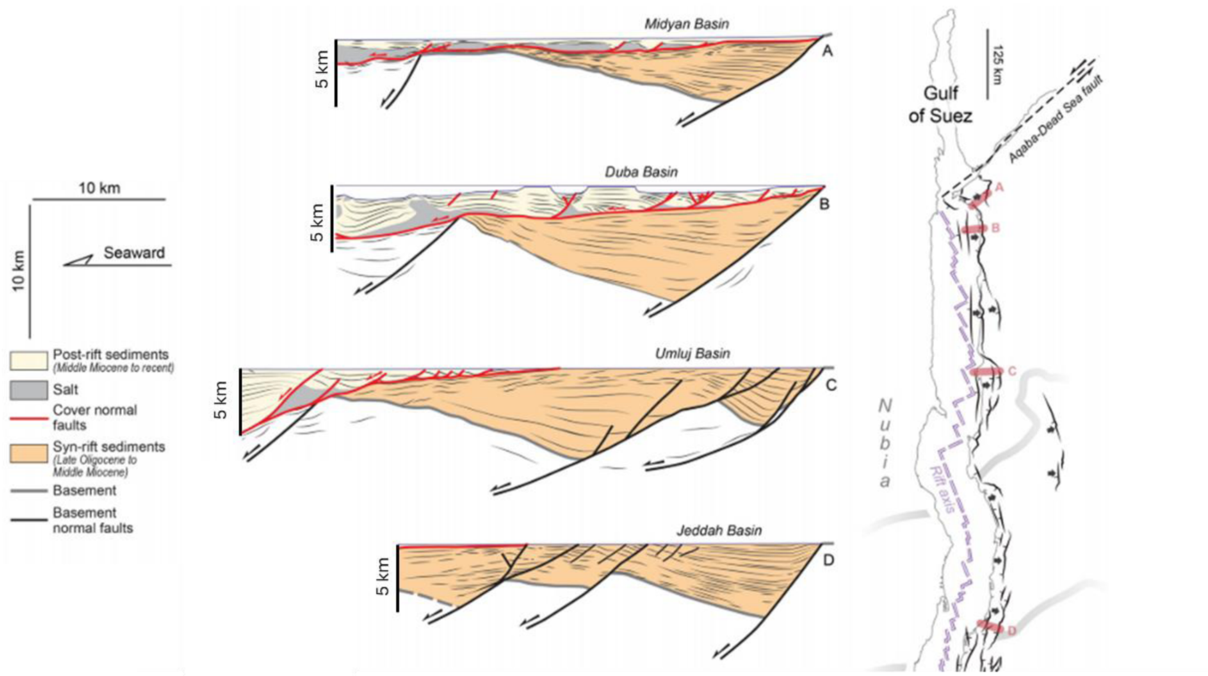

Figure 8—Typical West Coast rift basin architecture showing the syn-rift deposits, which are located at depths greater than 3 km, providing the conditions necessary to reach the required temperatures of over 120°C. Cross section (a) across Midyan Basin, (b) across Duba Basin, (c) across Umluj Basin, and (d) across Jeddah Basin. Modified from (Baby et al., 2024).

Lithium from Evaporative Brine Concentration. As seawater evaporates in a closed basin, lithium becomes concentrated in the remaining fluid while evaporite minerals begin to precipitate. Depending on the degree of evaporation, the concentration of lithium becomes significant. For this reason, lithium related to salt deposits can reach higher concentrations than lithium related, for example, to gypsum-anhydrite deposits.

On the west coast of Saudi Arabia, there are records of thick salt deposits starting from the Miocene. These salt deposits correspond to the Yanbu and Mansiyah Formations (Fig. 7). The Yanbu Formation marks the first appearance of restricted marine conditions in the early Miocene, though its occurrence is sporadic (Hughes & Johnson, 2005; Tubbs et al., 2014). In contrast, the Mansiyah Formation of late Middle Miocene age consists of massive evaporites with minor clastics. This formation represents a period of hydrological isolation and intense evaporation across the basin, leading to the deposition of a continuous layer of salt exceeding in places 1 km in thickness (Hughes & Johnson, 2005; Tubbs et al., 2014). The fluids associated with the precipitation of this thick layer of salt likely contain a significant amount of lithium.

Western vs Eastern Seaboard

The previous examples are from the rift basin on the west coast of Saudi Arabia. In this basin, there are sedimentary rocks that have sediments derived from the nearby Arabian Shield (Al Wajh Formation) buried at a sufficient depth to create the conditions necessary for Li-rich brines to be generated through water-rock interaction. That is, sediments containing Li-bearing minerals and temperatures greater than 120°C that facilitate the leaching of these minerals, releasing Li into the fluid. Additionally, the presence of thick Miocene salt deposits (Mansiyah Formation) suggests the potential of having lithium in brines derived from the remaining fluid linked to the precipitation of these salts.

On the other side of the plate, in the east coast Arabian Basin, sediments from the shield that were deposited during the Miocene or more recent periods, forming, for example, the Hofuf formation in the Eastern Province of Saudi Arabia (Alsafarjalani, 1999), are shallow and, therefore, do not reach the temperature needed for leaching lithium-bearing minerals. In addition, Hormuz salt deposits in the eastern parts of the Arabian Basin are Precambrian in age (Fig. 9). The associated evaporitic fluids may likely have been flushed from the pore waters, as the subsurface is dynamic and fluids tend to migrate. Another minor evaporite sequence present in the basin is the Jurassic gypsum/anhydrite of the Hith Formation (Fig. 9). The precipitation of this type of evaporite requires a lower degree of evaporation compared to halite, with significantly lower lithium content of the associated brines.

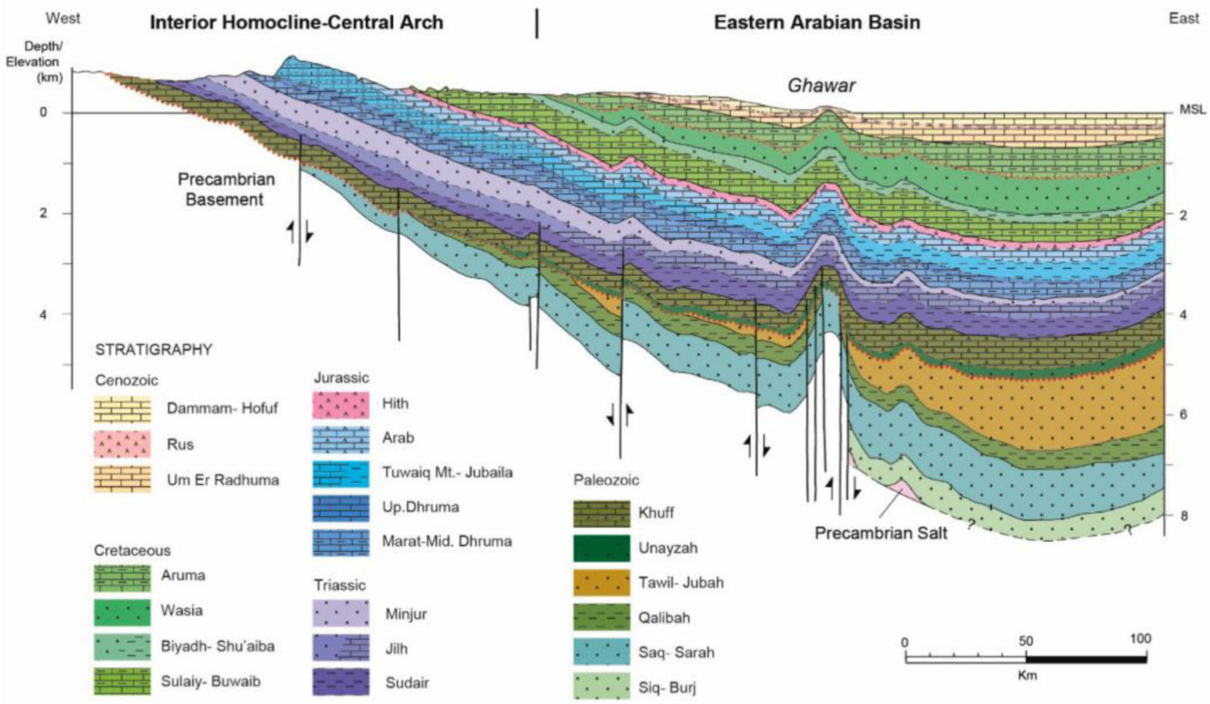

Figure 9—Geological cross-section of central and eastern Saudi Arabia. (From Ye et al., 2023)

Hence, based on these considerations and constrained by data, the deep brines of the west coast of Saudi Arabia have a significantly higher potential for a high lithium concentration compared to those from the east coast. Yet, lithium concentrations in the east coast Arabian Basin aquifers are still expected to be significantly increased compared to seawater.

Existing Data

Table 3 below shows the ICP-OES analysis results of three water samples taken in different places on the West Coast of Saudi Arabia. The first sample is water from the Red Sea. The Red Sea has higher lithium concentration than seawater from a normal marine environment; this is due to the arid conditions of the Red Sea, leading to higher evaporation and, therefore, higher concentration.

Table 3—ICP-OES results for some water samples in Saudi Arabia, concentration in ppm

Sample	*Li*	*Mg*	*Ca*	*Mg/Ca*	*Sr*	*Mn*	*Fe*	*Na*	*K*
Red Sea	*0.21 ±0.01*	*1487 ±6*	*410.9 ±1.3*	*3.62*	*4.78 ±0.02*	*< 0.01*	*< 0.01*	*12574 ±45*	*647.8 ±2.2*
KAUST 400m Well	*0.76 ±0.04*	*2376 ±34*	*8620 ±48*	*0.27*	*140.0 ±0.5*	*4.67 ±0.01*	*1.73 ±0.03*	*25238 ±15*	*359.3 ±0.8*
AL-Lith Hot Spring	*0.18 ±0.01*	*5.30 ±0.13*	*271.15 ±1.06*	*0.019*	*3.83 ±0.16*	*< 0.01*	*< 0.01*	*488.52 ±3.23*	*28.51 ±1.58*

The second sample is water from an exploratory geothermal well to a depth of 400 m on the campus of KAUST University. This sample has an elevated lithium content of three times that of the seawater. However, the Mg/Ca ratio is much reduced compared to that of seawater, possibly due to the incorporation of magnesium from the brine into diagenetic dolomite.

Al-Lith and Jizan are two known sites with hot springs in Saudi Arabia, located close to granitic rocks. However, the lithium content is reported to be low, up to 0.28 ppm for Al-Lith, and up to 0.78 ppm for Jizan (Al-Dayel, 1988; Chandrasekharam et al., 2015). A sample collected from a hot spring near Al-Lith confirms the low lithium concentration. The low lithium concentration in these fluids may result from limited water-rock interactions caused by relatively rapid fluid circulation at low temperatures similar to those measured in the hot springs of 70°C and slightly above 120°C (Al-Dayel, 1988). This may indicate that the hot springs are not linked to volcanism but are likely artesian, with recharge coming from meteoric water from the surrounding mountains.

Conclusions

This study presents the first geological assessment of Saudi Arabia's lithium potential, providing important insights for future exploration and sustainable resource development. The key conclusions are:

1. Direct Lithium Extraction (DLE) from geothermal brines offers a great alternative to conventional lithium production methods.
2. Lithium in subsurface brines comes from fluid-rock interaction with felsic igneous rocks and/or evaporative brine concentration. Volcano-sedimentary rocks formed by felsic rocks as detrital material are a potentially significant source of lithium in subsurface brines.
3. The west coast rift basins, which have syn-rift sedimentary formations with detrital material from the Arabian Shield, like the Al Wajh Formation, show strong potential for lithium-rich brines. High temperatures at depths greater than 3 km in these basins enable water–rock interactions that release lithium from minerals such as micas and volcanic glass into formation fluids. Additionally, thick Miocene salt deposits (e.g., Mansiyah Formation) suggest that concentrated lithium left over from the deposition of evaporites may also contribute to the lithium content of deep basinal aquifers, making these basins excellent candidates for Direct Lithium Extraction (DLE).
4. The eastern Arabian Basin is less favorable due to limited input of Li-bearing sediments from the Arabian Shield at appropriate depths, and the limited extent and old age of evaporites that could have contributed to the increase of lithium concentration in the remaining brine.
5. Geochemical data from a shallow well penetrating West Coast brines show increased lithium levels compared to seawater, likely due to evaporative brine concentration related to anhydrite formation.

Overall, this initial assessment identifies the west coast of Saudi Arabia as the most promising area for lithium exploration, offering an opportunity to support the Kingdom's efforts in advancing low-carbon technologies. However, there is a great need for more detailed exploration and sampling from deep wells and outcrops of the Arabian Shield to confirm and quantify Lithium resources.

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